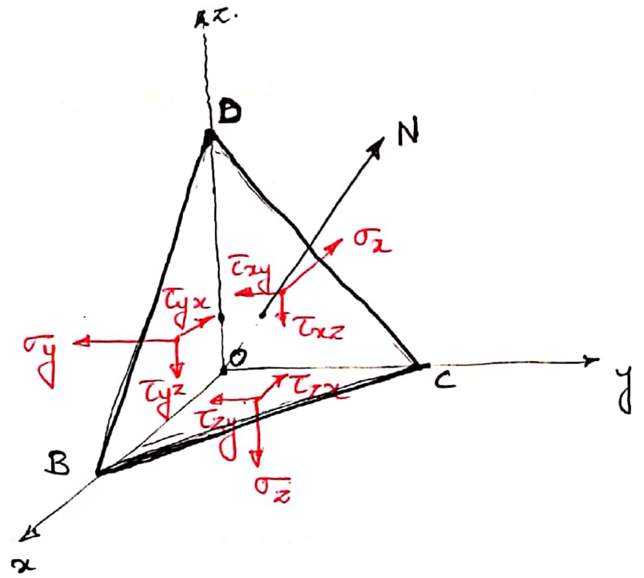
# Analysis of stress in 3D.

## Components of stress on a plane with direction cosines $(l, m, n)$.

Consider the tetrahedron of infinitesimal dimensions.




Tetrahedron is formed by the coordinate planes and a plane BCD with normal N.

The normal makes angles $\alpha, \beta, \gamma$ with $x$, $y$ and $z$ axes respectively.

The direction cosines of plane BCD

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma.$$

Let the area of the plane BCD be $A$. Then the area of the coordinate plane of the tetrahedron are

$Al$, $Am$ and $An$. (of the $x$, $y$ & $z$ planes).

The stresses acting on the coordinate planes are shown. There are

3 normal stresses : $\sigma_x, \sigma_y$ and $\sigma_z$

3 shearing stresses : $\tau_{xy} = \tau_{yx}$, $\tau_{xz} = \tau_{zx}$ and $\tau_{yz} = \tau_{zy}$.

Assumptions:

We assume that,

(1) The body forces can be neglected (it is of the order of $\epsilon^3$ where $\epsilon$ is the order of the ~~s~~ infinitesimal side of tetrahedron).

(2) Variation in stresses ~~across~~ over the sides ~~are~~ is also neglected. They are assumed to be uniform over the entire face.

~~Considering equilibr~~

Let ~~X be =~~ X, Y and Z be the stresses on the plane BCD along the coordinate axes. Then by considering the equilibrium of the element in the $x$ dir:

$$XA - \cancel{A}\,\sigma_x Al - \tau_{xy} Am - \tau_{xz} An = 0$$

$$\Rightarrow \boxed{X = \sigma_x l + \tau_{xy} m + \tau_{xz} n} \quad \text{——— (1)}$$

Similarly by considering equilibrium of the tetrahedron along the other two coordinate directions we get

$$\boxed{\begin{aligned} Y &= \tau_{xy} l + \sigma_y m + \tau_{yz} n. \\ Z &= \tau_{xz} l + \tau_{yz} m + \sigma_z n. \end{aligned}} \quad \text{——— (1)}$$

Thus, the components of stress on any plane defined by the direction cosines can be calculated from the above boxed eqns.

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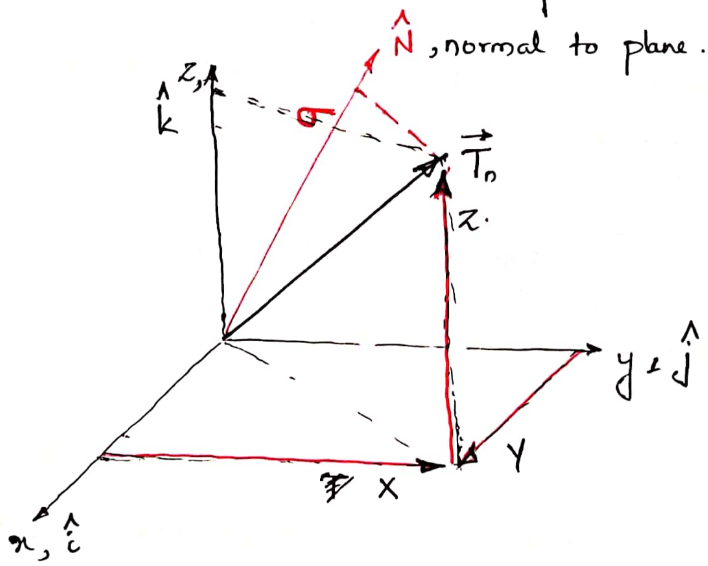
& Shearing

## Normal stresses on plane BCD.

$X, Y, Z$ are the components of the resultant ~~stress~~ traction vector $\vec{T}_n$ on the plane with normal $\hat{N}$.




$$\vec{T}_n = X\hat{i} + Y\hat{j} + Z\hat{k}.$$

The normal stress on the plane with normal $\hat{N}$ can be obtained from.

$$\sigma = \vec{T}_n \cdot \hat{N}. \qquad (\text{projection of } \vec{T}_n \text{ on } \hat{N}).$$

$$\sigma = X\underbrace{(\hat{N}\cdot\hat{i})}_{l} + Y\underbrace{(\hat{N}\cdot\hat{j})}_{m} + Z\underbrace{(\hat{N}\cdot\hat{k})}_{n}.$$

$$\Rightarrow \sigma = Xl + Ym + Zn. \quad \text{———} (2)$$

Substituting for $X, Y$ and $Z$ from (1) gives.

$$\sigma = (\sigma_x l + \tau_{xy} m + \tau_{xz} n) l + (\tau_{xy} l + \sigma_y m + \tau_{xz} n) m + (\tau_{xz} l + \tau_{xyz} m + \tau_{y} \sigma_z n) n.$$

$$\boxed{\sigma = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2 + 2\tau_{xy} lm + 2\tau_{yz} mn + 2\tau_{xz} ln} \quad \text{——} (3).$$

## Shearing stress on plane BCD

Knowing that $\sigma^2 + \tau^2 = |\vec{T}_n|^2$, where $\sigma$ is the normal stress and $\tau$ is the shearing stress, we can find that

$$\tau^2 = |\vec{T}_n|^2 - \sigma^2$$

$$\boxed{\tau^2 = (X^2 + Y^2 + Z^2) - \sigma^2} \quad (4)$$

where X, Y and Z can be obtained from Eq. (1) and $\sigma$ can be obtained from (3).

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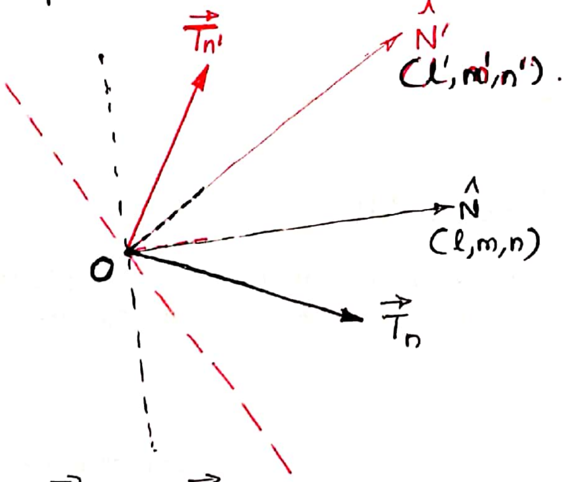
Consider two planes with normals $\hat{N}$ and $\hat{N}'$, passing through a point O. Let $\vec{T}_n$ be the resultant traction vector on plane $\hat{N}$ and $\vec{T}_{n'}$ be that on plane $\hat{N}'$.




We know that,

$$\vec{T}_n = X\hat{i} + Y\hat{j} + Z\hat{k}$$

$$\vec{T}_{n'} = X'\hat{i} + Y'\hat{j} + Z'\hat{k}$$

Note that, $\vec{T}_n$ and $\vec{T}_{n'}$ represent the same state of stress at point O, but at two different planes passing through O.

The projection of $\vec{T_n}$ on $\hat{N}'$

$$\vec{T_n}\cdot\hat{N}' = X(\hat{N}'\cdot\hat{i}) + Y(\hat{N}'\cdot\hat{j}) + Z(\hat{N}'\cdot\hat{k})$$

$$\vec{T_n}\cdot\hat{N}' = Xl' + Ym' + Zn'.$$

Similarly, the projection of $\vec{T_{n'}}$ on $\hat{N}$

$$\vec{T_{n'}}\cdot\hat{N} = X'l + Y'm + Z'n.$$

Substituting for X, Y and Z.

$$\vec{T_n}\cdot\hat{N}' = (\sigma_x l + \tau_{xy} m + \tau_{xz} n)l' + (\tau_{xy} l + \sigma_y m + \tau_{yz} n)m' + (\tau_{xz} l + \tau_{yz} m + \sigma_z n)n' \quad \text{———} (*)$$

Subst. for X', Y', Z'

$$\vec{T_{n'}}\cdot\hat{N} = (\sigma_x l' + \tau_{xy} m' + \tau_{xz} n')l + (\tau_{xy} l' + \sigma_y m' + \tau_{yz} n')m + (\tau_{xz} l' + \tau_{yz} m' + \sigma_z n')n \quad \text{———} (**)$$

From (*) and (**)

$$\boxed{\vec{T_n}\cdot\hat{N}' = \vec{T_{n'}}\cdot\hat{N}} \quad \text{———} (5)$$

# Principal Stresses.

We ~~Kss.~~ can ~~obtain~~ find the normal and shearing stresses on any plane from the ~~di~~ previous discussion, provided the state of stress at the given point is known. The state of stress can be expressed as 2nd order tensor (matrix).

$$[\sigma] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}.$$

The traction vector $\vec{T_n}$ on a given plane with normal $\hat{N}$ can be represented as a first order tensor (Vector).

$$\{T_n\} = \begin{Bmatrix} X \\ Y \\ Z \end{Bmatrix}.$$

And the normal vector $\hat{N}$ by

$$\{N\} = \begin{Bmatrix} l \\ m \\ n \end{Bmatrix}.$$

$\therefore$ Eq (1) on page (5) can be represented as.

$$\begin{Bmatrix} X \\ Y \\ Z \end{Bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \begin{Bmatrix} l \\ m \\ n \end{Bmatrix}. \quad \text{——— (6)}$$

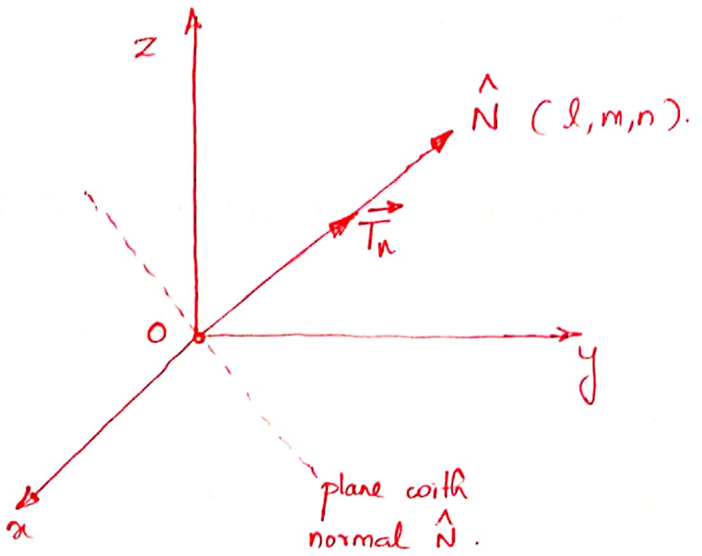
or

$$\boxed{\{T_n\} = [\sigma]\{N\}} \quad \text{——— (7)}$$

dir. cosine vector.

traction vector    stress matrix

Assume ~~that~~ ~~the~~ a plane with normal $\hat{N}$ on which ~~shearing stress is zero~~ the traction $\vec{T_n}$ is along the normal $\hat{N}$. Let the magnitude of traction be $\sigma$.




$$\Rightarrow \quad |\vec{T_n}| = \sigma. \quad \text{or} \quad \vec{T_n} = \sigma \hat{N}. \quad \text{———} (*)$$

Now, the components of the traction along the coordinate axes are.

$$X = \vec{T_n} \cdot \hat{i} = \sigma (\hat{N} \cdot \hat{i}) = \sigma l$$

$$Y = \vec{T_n} \cdot \hat{j} = \sigma (\hat{N} \cdot \hat{j}) = \sigma m$$

$$Z = \vec{T_n} \cdot \hat{k} = \sigma (\hat{N} \cdot \hat{k}) = \sigma n.$$

Now, substituting for X, Y and Z in (6) $\Rightarrow$

$$\begin{Bmatrix} \sigma l \\ \sigma m \\ \sigma n \end{Bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \begin{Bmatrix} l \\ m \\ n \end{Bmatrix}$$

$$\Rightarrow \begin{bmatrix} \sigma_x-\sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y-\sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z-\sigma \end{bmatrix} \begin{Bmatrix} l \\ m \\ n \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}. \quad \text{——— (8)}$$

Eq. (8) represents a set of homogeneous algebraic equations with variables $l, m, n$ (the dir. cosines of $\hat{N}$).

The equations are linearly dependent. Non-trivial sol$^n$s can be obtained by solving.

$$\begin{vmatrix} \sigma_x-\sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y-\sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z-\sigma \end{vmatrix} = 0.$$

$$\Rightarrow \sigma^3 - (\sigma_x+\sigma_y+\sigma_z)\sigma^2 + (\sigma_x\sigma_y + \sigma_y\sigma_z + \sigma_z\sigma_x - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2)\sigma$$
$$- (\sigma_x\sigma_y\sigma_z + 2\tau_{xy}\tau_{yz}\tau_{zx} - \sigma_x\tau_{yz}^2 - \sigma_y\tau_{xz}^2 - \sigma_z\tau_{xy}^2) = 0.$$

$$\Rightarrow \boxed{\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0.} \quad \text{——— (9)}$$

where $I_1 = \sigma_x + \sigma_y + \sigma_z$.

$$I_2 = \begin{vmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{vmatrix} + \begin{vmatrix} \sigma_y & \tau_{yz} \\ \tau_{yz} & \sigma_z \end{vmatrix} + \begin{vmatrix} \sigma_x & \tau_{xz} \\ \tau_{xz} & \sigma_z \end{vmatrix}$$

and $$I_3 = \begin{vmatrix} \sigma_x & \tau_{xy} & \tau_{zx} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{yz} & \sigma_z \end{vmatrix}$$

The three roots of the cubic eqn (9) are called the principal stresses. $(\sigma_1, \sigma_2, \sigma_3)$. The corresponding values of $l, m, n$ can be obtained by solving

* choose any 2 eqns

$$\left.\begin{aligned} (\sigma_x - \sigma_i) l + \tau_{xy} m + \tau_{xz} n &= 0 \\ \tau_{xy} l + (\sigma_y - \sigma_i) m + \tau_{yz} n &= 0 \\ \tau_{xz} l + \tau_{yz} m + (\sigma_z - \sigma_i) n &= 0 \\ \text{and} \quad l^2 + m^2 + n^2 &= 1. \end{aligned}\right] \text{———} (10)$$

* and this one

$i = 1, 2, 3$.

Each $\sigma_i$ is associated with a principal plane $\hat{N}_i$ with direction cosines $(l_i, m_i, n_i)$. On these planes, the traction vector is along the normal to the plane. Component of traction parallel to the plane is zero. Therefore shear stress ~~are~~ is zero along the principal planes.

## Stress Invariants

The ~~cub~~ roots of the cubic equation

$$\sigma^3 - I_1 \sigma^2 + I_2 \sigma - I_3 = 0 \text{ ———— } (9)$$

gives the principal stresses at the given point in the continuum. The principal stresses at a point depends only on the state of stress at that point. One may choose different frames of reference $(xyz)$ for representing the state of stress at a point. The principal stresses obtained from eqn (9) remains the same.

$\Rightarrow$ The coefficients of $\sigma^2$, $\sigma$ and the constant in eqn (9) should be the same irrespective of the frame of reference (xyz). Thus they (the coefficients) are called the Invariants (specifically, the invariants of stress tensor)

## Principal Planes are orthogonal to each other.

Choose two principal planes $\hat{N}$ and $\hat{N}'$ with the corresponding principal stresses $\sigma_1$ and $\sigma_2$.

The traction on each plane is given by (eqn (*) on page 9)

$$\vec{T}_n = \sigma_1 \hat{N} \quad \text{and} \quad \vec{T}_{n'} = \sigma_2 \hat{N}'.$$

We know that (eqn (5) on page (5))

$$\vec{T}_n \cdot \hat{N}' = \vec{T}_{n'} \cdot \hat{N}.$$

$$\Rightarrow \sigma_1 \hat{N} \cdot \hat{N}' = \sigma_2 \hat{N}' \cdot \hat{N}$$

$$\Rightarrow (\sigma_1 - \sigma_2)(\hat{N} \cdot \hat{N}') = 0. \quad \text{——— (11)}$$

In general $\sigma_1 \neq \sigma_2 \Rightarrow \hat{N} \cdot \hat{N}' = 0. \Rightarrow \hat{N} \perp \hat{N}'$

$\Rightarrow$ The principal planes are orthogonal to each other.

## Particular cases

(1) $\sigma_1 \neq \sigma_2 \neq \sigma_3$.

Each principal stress is associated with a unique principal direction. Let $\hat{N}_1, \hat{N}_2, \hat{N}_3$ be the corresponding principal directions. The ~~corresp~~ traction vector on the principal planes are given by (⊛ on page 9)

$$\vec{T}_{n_1} = \sigma_1 \hat{N}_1 , \quad \vec{T}_{n_2} = \sigma_2 \hat{N}_2 , \quad \vec{T}_{n_3} = \sigma_3 \hat{N}_3 .$$

Using eqn 5 on page 5

$$\vec{T}_{n_1} \cdot \hat{N}_2 = \vec{T}_{n_2} \cdot \hat{N}_1 .$$

$$\Rightarrow \sigma_1 (\hat{N}_1 \cdot \hat{N}_2) = \sigma_2 (\hat{N}_2 \cdot \hat{N}_1)$$

$$\Rightarrow (\sigma_1 - \sigma_2)(\hat{N}_1 \cdot \hat{N}_2) = 0$$

$$\Rightarrow \hat{N}_1 \cdot \hat{N}_2 = 0 \quad (\because \sigma_1 \neq \sigma_2).$$

$\hat{N}_1 \perp r . \hat{N}_2$

Similarly we get. $\hat{N}_2 \cdot \hat{N}_3 = 0 \quad (\because \sigma_2 \neq \sigma_3)$.

and $\hat{N}_1 \cdot \hat{N}_3 = 0 \quad (\because \sigma_1 \neq \sigma_3)$.

$\Rightarrow$ $\boxed{\hat{N}_1, \hat{N}_2 \text{ and } \hat{N}_3 \text{ are mutually perpendicular and unique.}}$

(2) $\sigma_1 = \sigma_2 \neq \sigma_3$.

Let the associated principal directions be $\hat{N}_1$, $\hat{N}_2$ & $\hat{N}_3$.

$$\vec{T}_{n_1} = \sigma_1 \hat{N}_1 \quad , \quad \vec{T}_{n_2} = \sigma_2 \hat{N}_2 \quad , \quad \vec{T}_{n_3} = \sigma_3 \hat{N}_3 .$$

$$\vec{T}_{n_1} \cdot \hat{N}_3 = \vec{T}_{n_3} \cdot \hat{N}_1$$

$$\Rightarrow (\sigma_1 - \sigma_3)(\hat{N}_1 \cdot \hat{N}_3) = 0$$

$$\Rightarrow \underline{\underline{\hat{N}_1 \perp r \ \hat{N}_3}} \qquad (\text{Since } \sigma_1 \neq \sigma_3).$$

Similarly $\vec{T}_{n_2} \cdot \hat{N}_3 = \vec{T}_{n_3} \cdot \hat{N}_2$.

$$\Rightarrow (\sigma_2 - \sigma_3)(\hat{N}_2 \cdot \hat{N}_3) = 0.$$

$$\Rightarrow \underline{\underline{\hat{N}_2 \perp r \ \hat{N}_3}} \qquad (\text{Since } \sigma_2 \neq \sigma_3)$$

And $\vec{T}_{n_1} \cdot \hat{N}_2 = \vec{T}_{n_2} \cdot \hat{N}_1$.

$$\Rightarrow \cancelto{0}{(\sigma_1 - \sigma_2)}(\hat{N}_1 \cdot \hat{N}_2) = 0.$$

$$\Rightarrow \hat{N}_1 \cdot \hat{N}_2 \in \mathbb{R} \text{ , a real value.}$$

- Thus we have a unique direction $\hat{N}_3$
- Any direction $\perp r$ to $\hat{N}_3$ is a principal direction

(Generally we choose two $\perp r$ directions ~~perpendicular~~ on the plane normal to $\hat{N}_3$).

(3) $\sigma_1 = \sigma_2 = \sigma_3$.

- $\vec{T}_{n_1} \cdot \hat{N}_2 = \vec{T}_{n_2} \cdot \hat{N}_1$.

$$\Rightarrow \cancel{(\sigma_1 - \sigma_2)}^{\to 0} (\hat{N}_1 \cdot \hat{N}_2) = 0$$

$$\Rightarrow \hat{N}_1 \cdot \hat{N}_2 \in \mathbb{R} \text{, a real value.}$$

- $\vec{T}_{n_2} \cdot \hat{N}_3 = \vec{T}_{n_3} \cdot \hat{N}_2$

$$\Rightarrow \cancel{(\sigma_2 - \sigma_3)}^{\to 0} (\hat{N}_2 \cdot \hat{N}_3) = 0$$

$$\Rightarrow \hat{N}_2 \cdot \hat{N}_3 \in \mathbb{R} \text{, a real value.}$$

- $\vec{T}_{n_1} \cdot \hat{N}_3 = \vec{T}_{n_3} \cdot \hat{N}_1$.

$$\Rightarrow \cancel{(\sigma_1 - \sigma_3)}^{\to 0} (\hat{N}_1 \cdot \hat{N}_3) = 0$$

$$\Rightarrow \hat{N}_1 \cdot \hat{N}_3 \in \mathbb{R} \text{, a real value.}$$

$\Rightarrow$ Every direction is a principal direction. This is the hydrostatic / isotropic state of stress.

(Generally we arbitrarily choose 3 mutually perpendicular directions as principal directions)